

# Hill Choy

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🌐 my website

## SUMMARY

AI Software Engineer experienced in deploying end-to-end machine learning and computer vision pipelines alongside robust backend systems. Adept at building complex Edge AI solutions, utilizing novel technologies to solve real-life challenges and automate physical workflows for commercial clients and local authorities.

## EDUCATION

### University College London (UCL)

MSc Computer Graphics, Vision and Imaging (**Current Average: 79%**)

Sept 2025 - Present

- **Highlighted Coursework:** Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

### Hong Kong University of Science and Technology (HKUST)

BEng in Computer Engineering (**Second Class Honors, Division 1**)

Sept 2020 - Jul 2024

- **Highlighted Coursework:** Intro to NLP, Computer & Communication Networks.

### IELTS Academic

8.0 (Reading: 8.5, Listening: 9.0, Writing: 7.0, Speaking: 7.0)

## RELEVANT EXPERIENCE

### Hyperspectral Image Analysis for Safe-Fabric Segmentation

United Kingdom

UCL, MSc Industry Dissertation Project (Partner: Zori Tex)

June 2026 – Sept 2026 (Expected)

- Collaborating with industry partner Zori Tex to research and develop an automated computer vision pipeline for textile waste sorting utilizing hyperspectral imaging.
- Project aims to explore PCA and shallow networks to condense high-dimensional data for memory-efficient CNN processing, alongside a weakly supervised Autoencoder approach to handle uneven real-world illumination.

### Research Assistant

Prof. Shueng-Han Gary Chan, CSE, HKUST

Hong Kong

Aug 2024 - Jun 2025

- **K11 MUSEA Cars and Pedestrian Flow Monitoring System**

- Engineered an end-to-end analytics system tracking 60,000+ daily visitors, utilizing OpenCV and a PyTorch-based YOLO model for real-time local flow counting on edge platforms.
- Developed a multithreaded Flask backend utilizing Redis for real-time data synchronization and SQLite/Pandas for robust storage, powering a web dashboard to optimize mall operations.

- **Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera**

- Contributed to the development of a privacy-preserving Edge AI fall-detection system deployed on an NPU-accelerated OrangePi, utilizing Human Pose Estimation for real-time patient monitoring.
- Developed rule-based post-processing algorithms to minimize false alarms without compromising true positive cases, achieving 98% accuracy and securing the Silver Award at the Hong Kong ICT Awards 2024.

- **Smart Traffic Management System (Kung Tong, Hong Kong)**

- Trained and deployed YOLO models on a dual-RTX 4090 server to process live streams from 33 cameras across 11 major junctions, monitoring 20,000+ daily vehicles per junction.
- Engineered automated client scripts using Python requests to securely transmit vehicle counts and yellow-box stop detections to a centralized API, handling Keycloak OAuth2 token authentication.

### Advanced Video Analytics and Edge AI for Smart Carpark Systems

Hong Kong

HKUST, Bachelor Degree Final Year Project (Supervisor: Prof. Shueng-Han Gary Chan, CSE, HKUST)

Sept 2023 - Apr 2024

- Designed an Edge AI smart car park system to track real-time space availability, utilizing OpenCV for video frame grabbing and preprocessing to handle complex feeds from custom low-light and wide-angle cameras.
- Deployed a PyTorch-based YOLOv8 model on a Raspberry Pi by integrating a MobileNetV3 backbone, converting to ONNX format, and utilizing Ultralytics for custom dataset annotation, halving inference latency and reducing parameters by 30% while maintaining baseline accuracy.

### Intern

Hong Kong Telecom (HKT), Integrated Service Provision Center

Hong Kong

June 2023 - April 2025

- Engineered a Python GUI executable to replace a legacy system and eliminate manual checks, automating both Excel and physical report generation for phone line data.
- Bypassed the lack of a native API using a custom web-scraping pipeline, dynamically extracting interface records to synchronize data with a MySQL database.